

SUMAKSHARIKA NAINAVARAPU

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PROFESSIONAL SUMMARY

Pharmacy-credentialed Data Analyst and consulting professional with 4+ years of combined experience in technical data analysis, data warehousing, ETL pipeline development, and advanced analytics across healthcare, specialty pharmacy, and financial services domains. Holds a B.S. in Pharmacy (Koneru Lakshmaiah University) and an M.S. in Health Informatics & Analytics (UNC Charlotte), delivering rare dual expertise in pharmacy operations, specialty drug lifecycle, and supply chain analytics. Skilled in SQL, Oracle/PL/SQL, Python, Power BI, Tableau, PostgreSQL, MySQL, and Snowflake. Proven track record designing executive dashboards, resolving data discrepancies, managing data governance, supporting data warehouse initiatives, and delivering actionable insights for operational and executive audiences. Experienced in SP operations, SP data reporting, formulary management, prior authorization tracking, and drug utilization analytics aligned with specialty pharmacy consulting engagements.

TECHNICAL SKILLS

Languages & Querying: Python (Pandas, NumPy), SQL, Oracle/PL/SQL, PostgreSQL, MySQL, Snowflake, Advanced SQL (CTEs, Window Functions)

Databases & Warehousing: Enterprise Data Warehouses, Data Modeling (Fact & Dimension Tables, Star Schema), ER Modeling, Data Normalization

BI & Reporting Tools: Power BI, Tableau, SQL Developer, Visual Studio (familiar), Streamlit, Qlik (familiar), Executive Dashboards, KPI Reporting

Data Engineering & ETL: ETL Pipelines, Data Extraction/Transformation/Loading, Data Validation, Data Reconciliation, Data Quality Checks, Structured/Semi-structured/Unstructured Data Formats

Analytics & Modeling: EDA, Cohort Analysis, Time-Series Analysis, Logistic Regression, Risk Stratification, Feature Engineering, Statistical Analysis, XGBoost, scikit-learn, MLflow, HuggingFace/FinBERT, ARIMA, Prophet

Healthcare & Pharmacy: Specialty Pharmacy Operations, SP Data Reporting, Drug Lifecycle Management, Pharmacy Supply Chain, Formulary Management, Prior Authorization, Charlson Comorbidity Index, EHR Data Analysis, Readmission & LOS Analysis

Domain Expertise: Specialty Pharmacy Consulting, Financial Services, Banking & Financial Services Analytics, Healthcare & Clinical Analytics

Governance & Collaboration: Data Governance, Metadata Management, Audit Readiness, Regulatory Compliance, Confluence, Jira, RCA, GitHub, Anaconda, MS Excel

PROFESSIONAL EXPERIENCE

Data Analyst | HSBC | Remote, USA | 10/2024 – Present

- Analyzed large-scale customer, transaction, and event-level digital banking datasets using SQL and Python to identify usage patterns, behavioral trends, funnel metrics, and engagement KPIs — supporting data-driven decision-making across business units.
- Partnered with business stakeholders, product managers, technology teams, and operations leadership to gather analytics requirements and translate business needs into scalable data solutions aligned with organizational and regulatory objectives.
- Extracted, cleaned, and transformed structured and semi-structured data from Snowflake enterprise data warehouses using advanced SQL (CTEs, window functions) and Python (Pandas, NumPy), improving data accuracy and consistency by 22% through automated data validation, reconciliation, and quality checks.
- Designed, developed, and maintained interactive Tableau dashboards and self-service BI reports integrating data from Snowflake and internal systems to monitor KPIs, customer engagement, funnel conversion, and operational performance for executive and operational audiences.
- Managed reporting environment including data sources, metadata definitions, business logic documentation, and access governance standards, ensuring compliance with internal data governance and risk frameworks.
- Performed QA and validation of imported, transformed, and reported data across multiple source systems, assessing downstream impact of source production system changes on reporting and analytics pipelines.
- Assessed and validated the impact of changes to source production systems on downstream reporting and analytics pipelines, mitigating risk of data inconsistencies across 15+ enterprise reports.
- Delivered data-driven recommendations contributing to a 20% improvement in customer follow-up efficiency, 14% improvement in customer satisfaction scores, and 35% increase in digital banking adoption.
- Documented data sources, business definitions, methodologies, and analytical findings in Confluence, ensuring transparency, reproducibility, and audit readiness aligned with regulatory compliance requirements.

Data Analyst | Tata Consultancy Services (TCS) | Hyderabad, India | 01/2021 – 08/2023

- Delivered consulting engagements for healthcare clients, collaborating with clinical SMEs, business analysts, product owners, and IT teams to gather requirements, define KPIs, and translate operational needs into scalable data analytics solutions.
- Extracted, transformed, and validated large-scale EHR and pharmacy datasets using SQL and Python (Pandas, NumPy), improving data quality and reducing missing or inconsistent records by 18% through automated validation and reconciliation pipelines.
- Developed and implemented risk stratification models using logistic regression and the Charlson Comorbidity Index (CCI) to identify high-risk patients, supporting early intervention strategies and contributing to a 22% reduction in 30-day readmissions.
- Designed and delivered interactive Power BI dashboards and analytical reports using MS Excel (PivotTables, advanced formulas, data validation) to visualize KPIs, trends, variance analysis, and performance metrics for executive leadership and client stakeholders.
- Automated recurring reports and ETL pipelines from multiple source systems, reducing manual reporting effort and enabling near real-time insights for senior stakeholders and consulting clients.
- Supported data governance initiatives including metadata management, data normalization, and standardization across clinical systems, commissioned and decommissioned datasets in alignment with data lifecycle and governance requirements.
- Documented ETL processes, data models, and business logic in Confluence, ensuring audit readiness, regulatory compliance, and long-term maintainability across all reporting environments.
- Monitored KPIs using control charts, trend analysis, variance analysis, and RCA/PDSA cycles to support continuous improvement initiatives across cardiology and orthopedic departments.

KEY PROJECTS

Pharmacy Supply Chain & Prescription Analytics | PostgreSQL, Oracle-Compatible SQL, Python ETL, Streamlit

- Built end-to-end pharmacy supply chain platform covering drug inventory management, supplier performance scoring, prescription fill rates (8,000 Rx), stockout analysis, and automated reorder alerts. Oracle-compatible SQL schema with stored function fn_reorder_recommendation and inventory reconciliation ETL.

Specialty Pharmacy Claims Analytic | PostgreSQL, Python ETL, PL/pgSQL, Streamlit

- Developed SP claims pipeline on 2,000 specialty pharmacy claims across 10 drugs and 7 payers: full PostgreSQL schema with ER model, PL/pgSQL audit triggers, 11 KPI SQL queries, and 5-tab executive Streamlit dashboard covering denial analysis, dispensing SLA, and payer mix.

Drug Utilization & Formulary Analytics | MySQL, Python ETL, Stored Procedure, Streamlit

- Built formulary tier management and drug utilization analytics on 5,000 Rx across 20 specialty and non-specialty drugs; MySQL stored procedure sp_formulary_tier_impact(), 2,047 prior authorization requests analyzed, and generic substitution opportunity engine.

Healthcare Data Quality & Governance Pipeline | PostgreSQL, Python Rules Engine, Streamlit

- Engineered configurable DQ rules engine with 21 validation rules across 4 clinical datasets; severity-weighted scoring, data profiling, audit trail, reconciliation reporting, and downstream impact registry tracking 7 reports — 998 issues caught and flagged for remediation.

Pharmacy Readmission Risk Predictor | XGBoost, scikit-learn, MLflow, Charlson Comorbidity Index, Streamlit

- Built 30-day readmission risk prediction on 2,500 patients using CCI ICD-10 scoring, pharmacy features, and lab values; Logistic Regression ROC-AUC 0.8432; MLflow experiment tracking; interactive CCI calculator for real-time risk scoring.

EDUCATION

M.S. in Health Informatics & Analytics | University of North Carolina at Charlotte | 08/2023 – 05/2025

B.S. in Pharmacy | Koneru Lakshmaiah University, India | Pharmacy Operations · Drug Lifecycle · Clinical Data | 08/2018 – 05/2022

CERTIFICATIONS

Tableau Desktop Specialist | Tableau

Data Analyst Track | DataCamp

Google Advanced Data Analytics Certificate | Coursera (In Progress)